Phrase 10

Table 1. Dual-Stream Model Evaluation Results

Metric	Value
Accuracy	0.8541
Precision	0.8483
Recall	0.8625
F1-Score	0.8554
Specificity	0.8458
AUC-ROC	0.9305

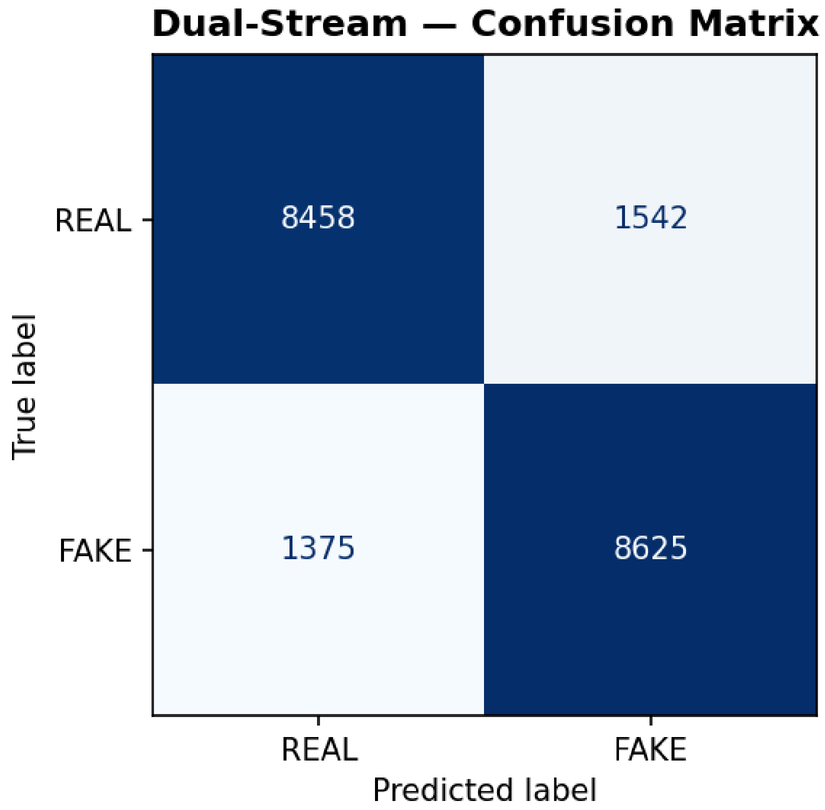
Table 2. Confusion Matrix

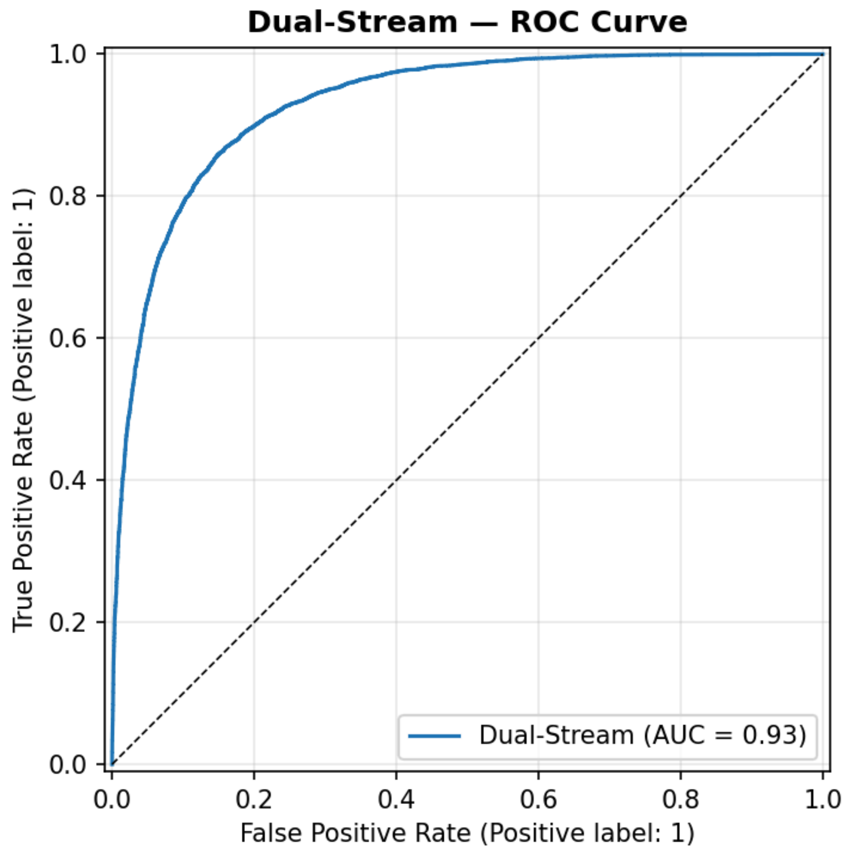
	Predicted FAKE	Predicted REAL
Actual FAKE	**8,625 (TP)**	**1,375 (FN)**
Actual REAL	**1,542 (FP)**	**8,458 (TN)**

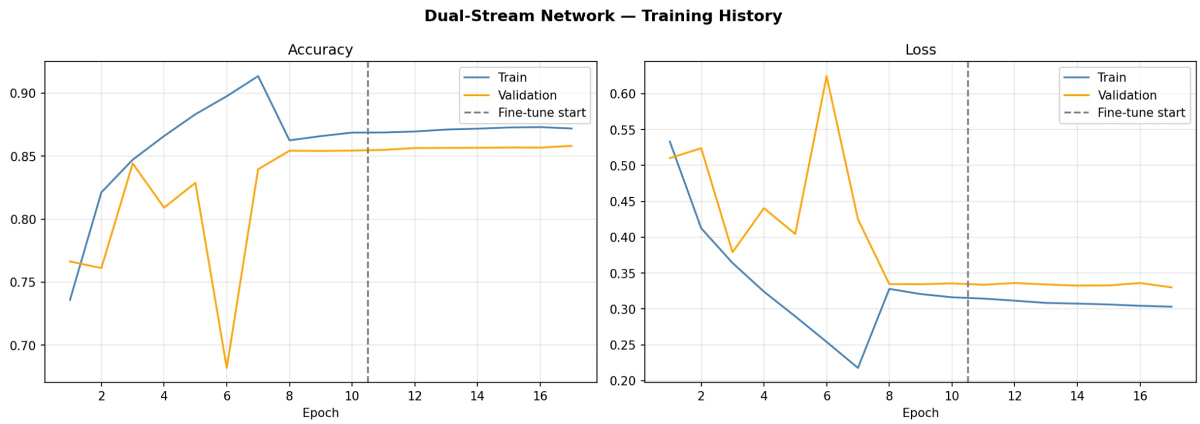
Table 3. Evaluation Summary

Observation	Interpretation
Accuracy	The model correctly classifies **85.41%** of all test images.
Precision	A precision of **84.83%** indicates that most images predicted as fake are correctly identified.
Recall	A recall of **86.25%** demonstrates strong detection of fake images with relatively few missed cases.
F1-Score	The F1-score of **85.54%** reflects a well-balanced trade-off between precision and recall.
Specificity	The model correctly identifies **84.58%** of real images, indicating balanced performance across both classes.

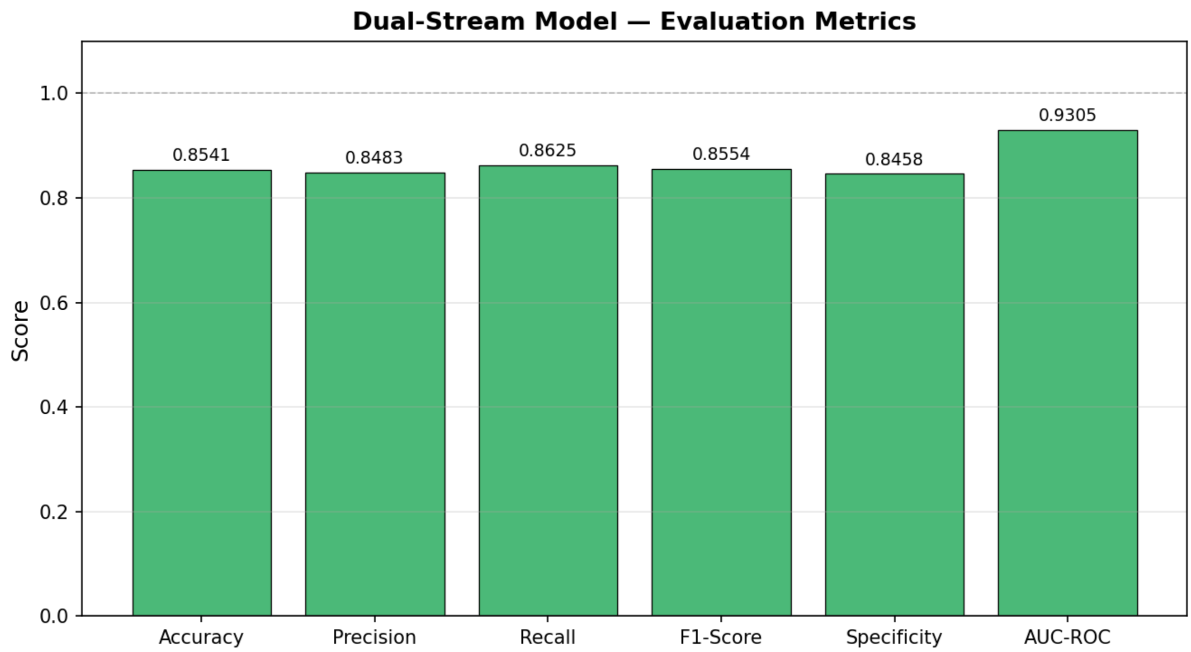
AUC-ROC	An AUC-ROC of **0.9305** demonstrates excellent discrimination between real and fake images.
Key Finding	Combining spatial and frequency-domain features in a dual-stream architecture yields more balanced and robust deepfake detection than using either feature type alone.


Dual-Stream — Confusion Matrix
True label
REAL
FAKE
8458
1542
1375
8625
REAL
FAKE
Predicted label


Dual-Stream — ROC Curve
True Positive Rate (Positive label: 1)
False Positive Rate (Positive label: 1)
0.0
0.2
0.4
0.6
0.8
1.0
Dual-Stream (AUC = 0.93)



Dual-Stream Network — Training History
Accuracy
Loss
Train
Validation
Fine-tune start
Epoch



Dual-Stream Model — Evaluation Metrics
Score
0.8541
0.8483
0.8625
0.8554
0.8458
0.9305
Accuracy
Precision
Recall
F1-Score
Specificity
AUC-ROC